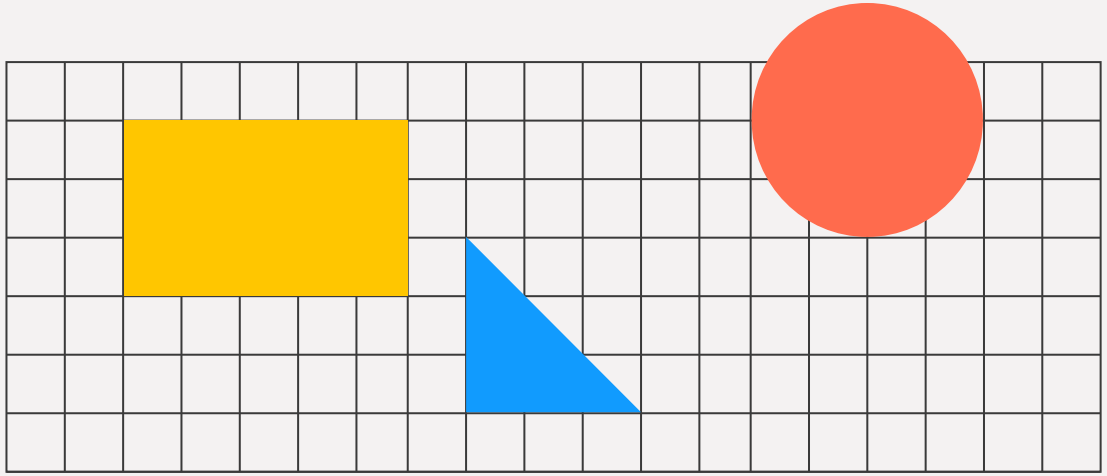
A2: POVs, HMWs, Experience Prototypes



The Team



Ryan L.

Computer Science ‘28



Brian L.

Bioengineering ‘26
Computer Science ‘27



Cherie C.

Data Science ‘27

Additional Needfinding

Creativity in the age of AI

Creativity in the age of AI for those who are too busy to practice it

Scope of Project

From A1, we found...

Clear

- Time is the real barrier, more than motivation or skill
- Consistency beats intensity; skill grows from small, regular practice, not bursts
- Effort is the point: people want to make things themselves, not shortcut the entire process
- Once it feels like an obligation, it stops being enjoyable

Unclear

- When do people actually get free time, and what do they do with it instead?
- We wanted to narrow down our intended audience/user-base

Narrowing our target population from A1, we were looking for busy individuals with **creative hobbies but no bandwidth**

Our interviewees: techies with creative hobbies



Edward

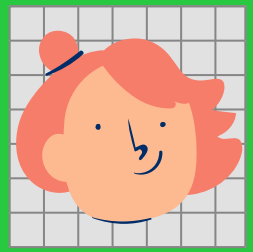
Senior @ UBC
Systems Engineering Intern @ Rivian & Volkswagen
Artist



Harold

AI Engineer @ Meta
Singer-songwriter

*Extreme



Hunter

Stanford Alumnus
Senior SWE @ Google
Two-time published author

*Extreme

Edward: Empathy Map

Says

Thinks

Does

Feels



Edward

Senior @ UBC
Systems Engineering Intern @
Rivian & Volkswagen

Edward: Empathy Map

Says

"If my mind's melting, I'll just scribble some random stuff on a notebook and then go back to work"

Brian

Thinks

Today you can only succeed online by conforming to formats or already having an audience

Brian

Does

Started a 100-day drawing challenge and stopped at day 40 when university began

Brian

Feels

Feels discouraged when his posts stalled just as his skills plateaued

Brian



Edward

Senior @ UBC
Systems Engineering Intern @ Rivian & Volkswagen

Art is Edward’s escape from burnout, but in his relentless pursuit of perfection, he **burnt out again**

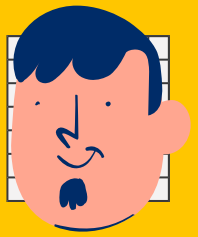
Harold: Empathy Map

Says

Thinks

Does

Feels



Harold

AI Engineer @ Meta
singer-songwriter

Harold: Empathy Map

Says

"But what differentiates a song that you would want to listen to again from a song that you're like, okay, yeah, that was neat, is like a lot of the human behind it."

Ryan

Thinks

Consistent progress is preferable to bursts of progress, but sometimes it's difficult to stay consistent with life because work can get very busy

Ryan

Does

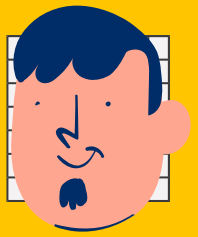
Clocks out of work around 6-7 pm and oftentimes spends the rest of his evening doing chores/ going on dates with his wife

Ryan

Feels

A sense of accomplishment when he finishes a song that is "the best feeling in the world"

Ryan



Harold

AI Engineer @ Meta
singer-songwriter

As a working professional, there is **not enough time** in the day to work, tend to your family, and pursue a creative hobby

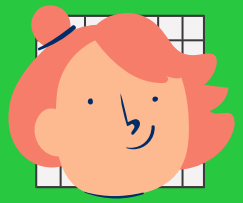
Hunter: Empathy Map

Says

Thinks

Does

Feels



Hunter

Stanford Alumnus
Senior SWE @ Google
Two-time published author

Hunter: Empathy Map

Says

"Whenever I see something artistic that I like, I'm always thinking, what would it take to make that?"

Brian

Thinks

Human creativity needs protecting, in themselves and in others

Brian

Does

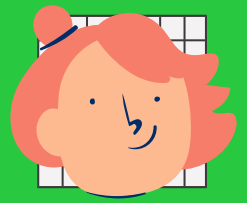
Published a fantasy duology; a sci-fi novel is coming in early 2028

Brian

Feels

Fiercely protective of human creativity, in themself and others

Brian



Hunter

Stanford Alumnus
Senior SWE @ Google
Two-time published author

Inspiration is a derivative of human creativity, but a creative's inspiration is **limited by their time**

POV Development



POV: Christina

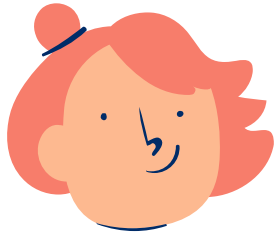
We met Christina, a Stanford graduate student in HCI and former Google and Disney UX intern who loves building every frame by hand in Figma.

We were surprised to notice Christina's solution to a summer "crisis" over **"why is all I do is work?"** wasn't changing jobs but buying a pocket camera she now carries everywhere; a small, hands-on practice that is far from the "money and numbers and how many clicks" that measured the value of her work.

We wonder if this means a creative practice only restores what work drains if it stays free of the speed and the measurement that drained it.

It would be game-changing to help Christina with demanding work keep a **small, hands-on creative practice** alive **without it becoming another metric or deadline**

Key HMWs for Christina



How might we make a creative practice feel like tending a garden rather than closing a ticket?

How might we keep a creative practice a refuge from demanding work, instead of becoming more of it?

How might we keep making by hand meaningful when AI can produce the same output instantly?

POV: Harold

We met Harold, a 32-year old AI engineer at Meta who pursues songwriting as a creative outlet and stress relief and has been since high school

We were surprised to notice that Harold, rather than songwriting consistently, songwrites in an annual week-long in-person spurt with his best friend since his **free time is inconsistent** relative to his relatively volatile work and family life.

We wonder if this means Harold feels an obligation to spending personal time with his wife rather than his hobbies and therefore he **lacks consistency in pursuing them.**

It would be game-changing if we could help Harold find time throughout the week to **meaningfully work on and share his creative hobbies** without sacrificing time with his wife.

Extreme User:

Full time software engineer (busy career, technical, non-creative in the traditional sense) who deeply engages in creativity via singing songwriting but with a self-inflicted inconsistency

Name Department Course Number Section Time Location DATE

Key HMWs for Harold



How might we shorten Harold's songwriting process so he can fit it into his nighttime routine?

How might we connect Harold with other people interested in songwriting, such as his friend, to inspire him to create more consistently?

How might we help Harold find less time consuming creative hobbies that might also be a source of creative outlet and stress relief?

POV: Hunter

We met Hunter, a Stanford CS alumnus and senior software engineer at Google with a deep passion for music and creative writing. They have **published a science fiction and fantasy duology.**

We were surprised to notice how willing Hunter was to try new hobbies and learn new skills for the sake of their creative practice, from learning to ride a motorcycle to **starting a metal band.**

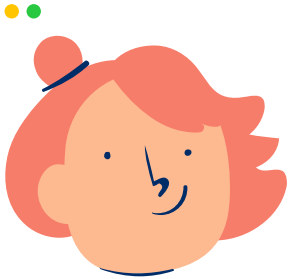
We wonder if this means Hunter, who has fit themself into the role of a software engineer over the years, was always drawn to their bolder hobbies, or whether world building and story writing simply became the reason to pursue them.

It would be game-changing if Hunter could easily find **guidance** and access to a diverse range of hobbies that fit the context and inspiration of their creative pursuits.

Extreme User:

Full time software engineer (busy career, technical, non-creative in the traditional sense) who has managed to publish a duology (consistent schedule and creative practice)

Key HMWs for Hunter



How might we lower the cost, time, and intimidation of trying an unfamiliar hobby as a total beginner?

How might we help people turn the questions their creative work raises into experiences they can live firsthand?

How might we help someone who's "oversubscribed" take on a new pursuit without dropping an old one?

Top 3 HMWs

Christine's POV

1

How might we keep a creative practice a **refuge** from demanding work, instead of becoming more of it?

Hunter's POV

2

How might we turn the questions people's creative work raises into experiences they can live **firsthand?**

Hunter's POV

3

How might we lower the cost, time, and intimidation of trying an unfamiliar hobby as a **total beginner?**

Solutions



Solution 1

How might we keep a creative practice a **refuge** from demanding work, instead of becoming more of it?



UGH-ly Hour: prompts that ask for deliberately ‘meh’ or silly art, to keep pressure away

UGH-ly Hour: prompts that ask for deliberately ‘meh’ or silly art, to keep pressure away

How will this solution respect the time and attention of hobbyists while also addressing their perfectionism?

We are assuming that hobbies can be demanding for everyone.

Assumption Map

unknown

Not vital to the user experience

Vital to the user experience

known



UGH-ly Hour

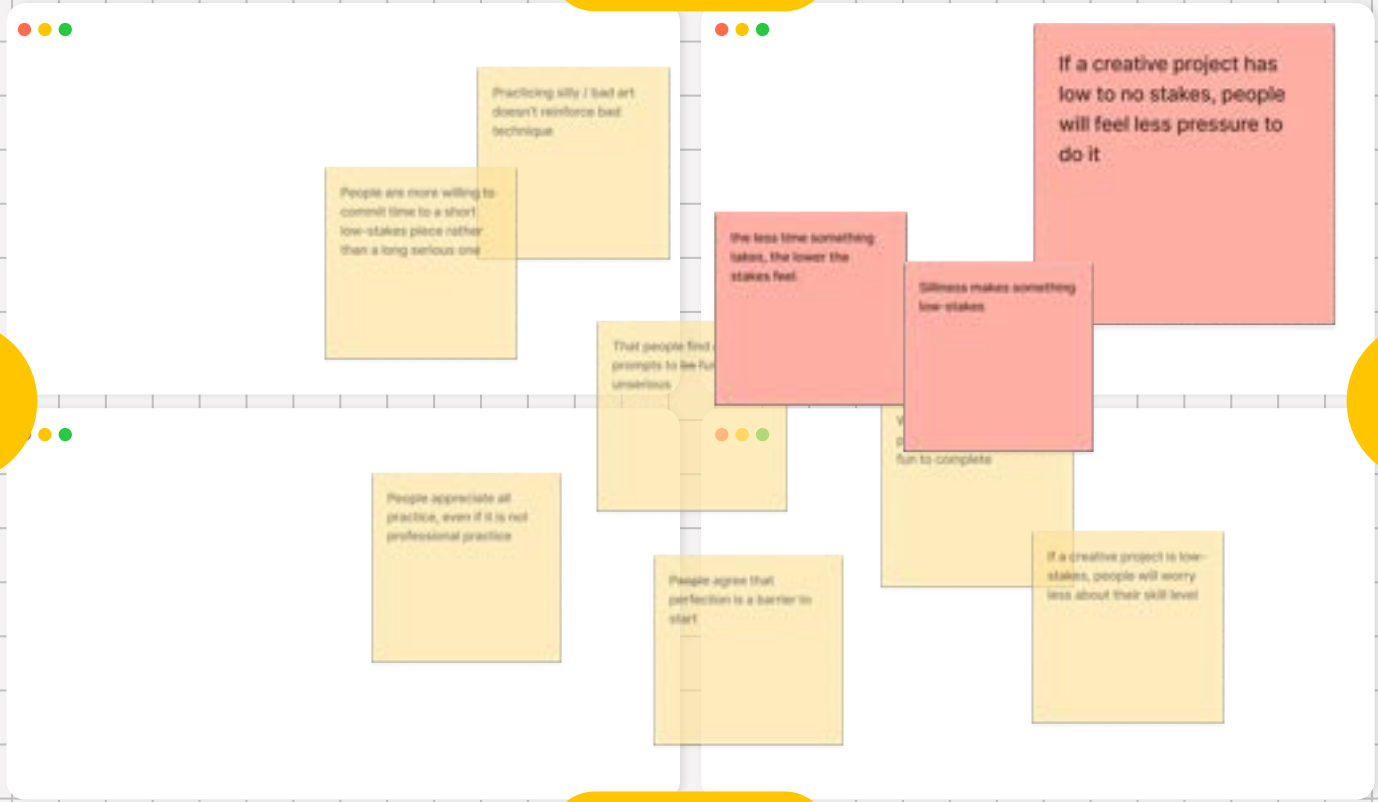
Assumption Map

unknown

Not vital to the user experience

Vital to the user experience

known



UGH-ly Hour

Solution 2

How might we turn the questions people's creative work raises into experiences they can live **firsthand?**



Create **immersive, multisensory moodboards** to understand an experience enough to base creative work off of it

Create **immersive, multisensory moodboards** to understand an experience enough to base creative work off of it

How will this solution serve people who don’t have certain senses, like sight or hearing?

Moodboards require the art and work of other people: how to respect and credit them?

Assumption Map

unknown

Not vital to the user experience

Vital to the user experience

known

People would rather use a generated/community-made moodboard over a moodboard they create themselves

People would rather experience things digitally rather than physically engaging themself in that experiential space

Immersion is a fair substitute for a sociophysical experience

People would prefer a moodboard using more senses than just sight

Adding more senses into the moodboard will give the user an immersive experience → user can gain more inspiration

People might not always have the bandwidth to go outside and complete an experience

If the creative work involves unrealistic ideas (fantasy elements), that "experience" might not exist

Moodboard

Assumption Map

unknown

People would prefer a moodboard using more senses than just sight

Ryan

Adding more senses into the moodboard will give the user an immersive experience → user can gain more inspiration

Brian

People would rather use a generated/community-made moodboard over a moodboard they create themselves

People would rather experience things digitally rather than physically engaging themself in that experiential space

Not vital to the user experience

Vital to the user experience

People might not always have the bandwidth to go outside and complete an experience

If the creative work involves unrealistic ideas (fantasy elements), that "experience" might not exist

known

Moodboard

Solution 3

How might we lower the cost, time, and intimidation of trying an unfamiliar hobby as a total beginner?



Piggybank: making space where nothing can be saved, screenshot, shared, or posted until user fulfills a quota

Piggybank: making space where nothing can be saved, screenshot, shared, or posted until user fulfills a quota

For people with highly variable executive dysfunction, how do our motivation principles come into conflict with theirs?

This is primarily a solution for people who have a non-zero amount of consistent free time

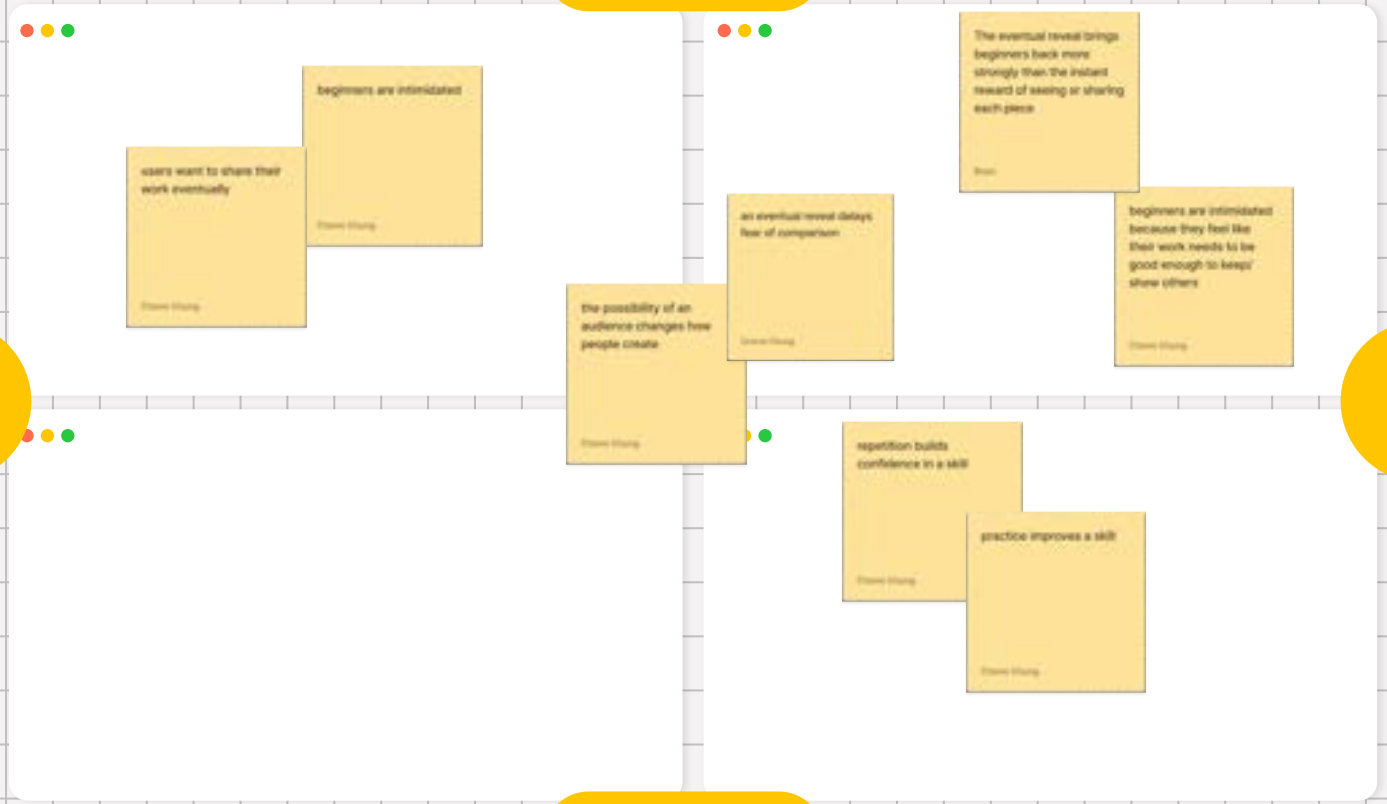
Assumption Map

unknown

Not vital to the user experience

Vital to the user experience

known



Piggybank

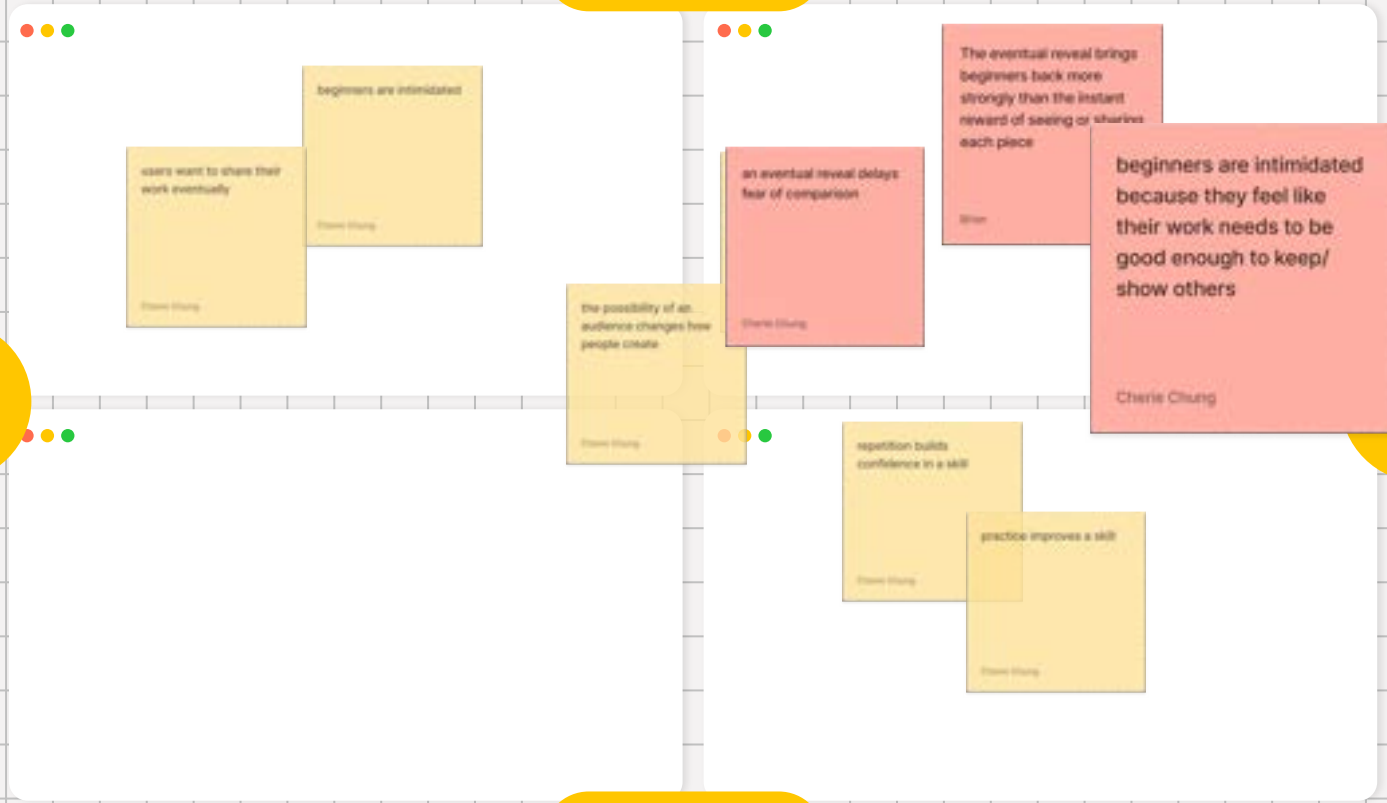
Assumption Map

unknown

Not vital to the user experience

Vital to the user experience

known



Piggybank



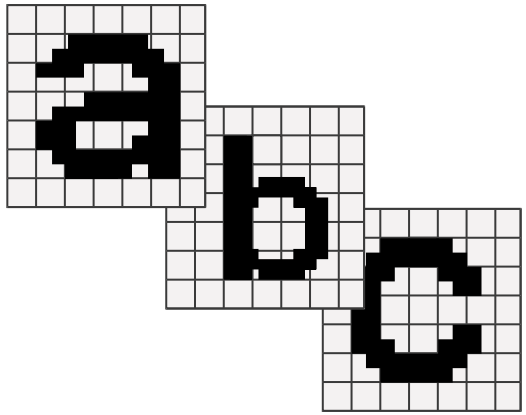
Experience Prototypes

UGH-ly Hour

Prototype #1

Critical Assumption:
If a creative project has low to no stakes, people will feel less pressure to do it

If silliness doesn't lower pressure, this is just another task, which is what HMW 1 is trying to avoid.



Participant: Constance



- **4th year CS student @ Stanford**
- **Used to draw as a hobby, but has drawn less recently because of school and other responsibilities.**

Experience Prototype #1

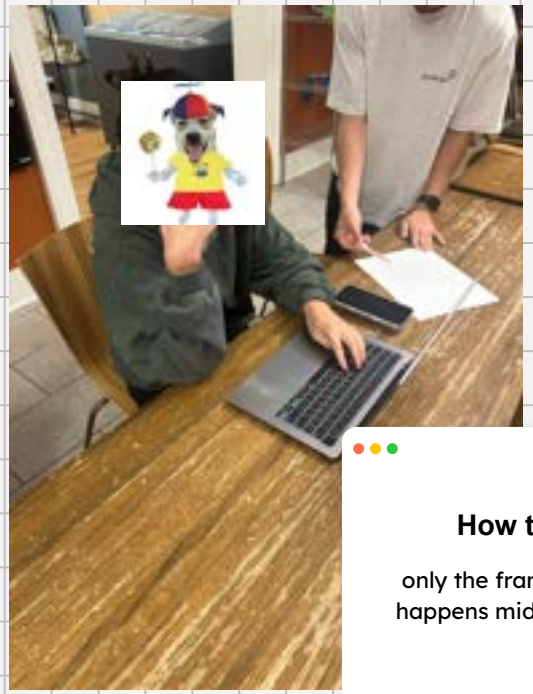
Setup:

The participant does real work. A teammate texts **two 3-minute "creative break"** prompts with the same time limit and materials.

Normal: "draw a self-portrait."

Silly: "draw your favorite meme."

After each, a paper slip **rates 1–5** how much it mattered and how much they judged themselves, then the drawing goes in a KEEP or TOSS bucket.



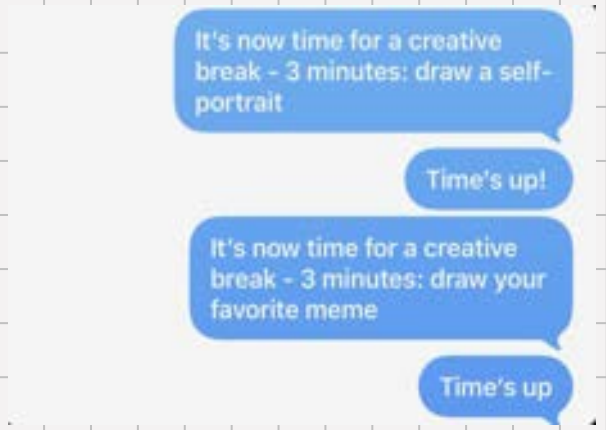
How this tests the assumption:

only the framing changes between rounds, and it happens mid-work, where a refuge has to hold up.

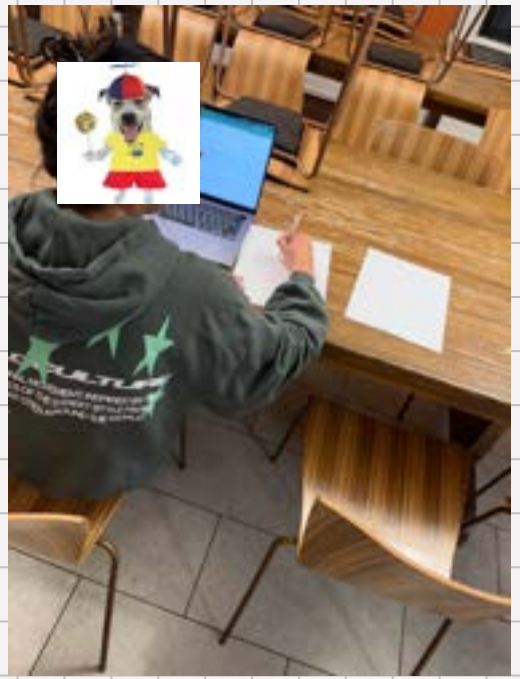
1. Working hard



2. Prompt Texted



3. Creative Break!



Prototype 1 Creations

1st Prompt
5 minutes in



2nd Prompt
13 minutes in



	Normal: Self-Portrait	Silly: Favorite Meme
How much the drawing mattered (1-5)*:	2	4
How much they judged their drawing (1-5)*:	4	2
Keep or Toss	Toss	Keep

* 1 = not at all, 5 = sooo much

Name Department Course Number Section Time Location DATE

Assumption Reflections

What Worked

- **Less self-judgment:** 2 silly vs. 4 serious
- **Valued more:** kept the dog, tossed the portrait
- **More fun:** “I had more fun doing the dog”
- **Time cap helped:** 3 min “made it feel more approachable”

What Didn’t Work

- **Serious prompt froze them:** "What would a creative self portrait be?"
- **3 min felt long** mid-work
- **Prompts differed in subject,** not only silliness
- **One participant,** order not recorded

Insights

Did assumption hold?

If a creative project has low to no stakes, people will feel less pressure to do it

= Held

Insights:

1. **Silly lowered judgment, not value:** judged 2 vs. 4, mattered 4 vs. 2, and it's the one they kept.
2. **The time cap does half the work:** "approachable" because it was only 3 minutes.
3. **Serious, open prompts read as a test:** "What would a creative self portrait be?"

Going forward: concrete silly prompts, under 3 minutes, nothing shared.

Immersive Moodboard

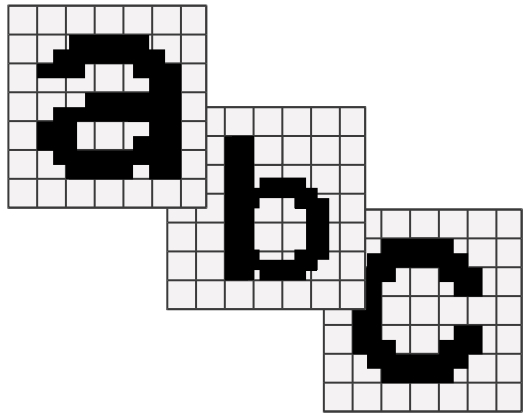
Prototype #2

Critical Assumptions:

Adding senses to a moodboard will make it more immersive

Immersion == inspiration

People would prefer engaging multiple senses over traditional visual moodboarding



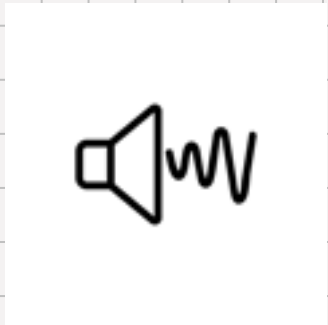
Experience Prototype #2

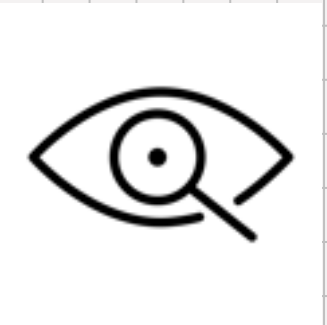
Setup:

The participant was tasked to imagine “What do you imagine being in a band is like?” (role play) and **moodboard in silence for 5 minutes** on a blank Google Slide

Then, the participant **chooses music** that reminds them of the prompt and **moodboards** on another blank slide **for another 5 minutes**

The participant is then prompted for feedback and explaining their process with no-music versus with-music moodboarding





How this tests the assumption:

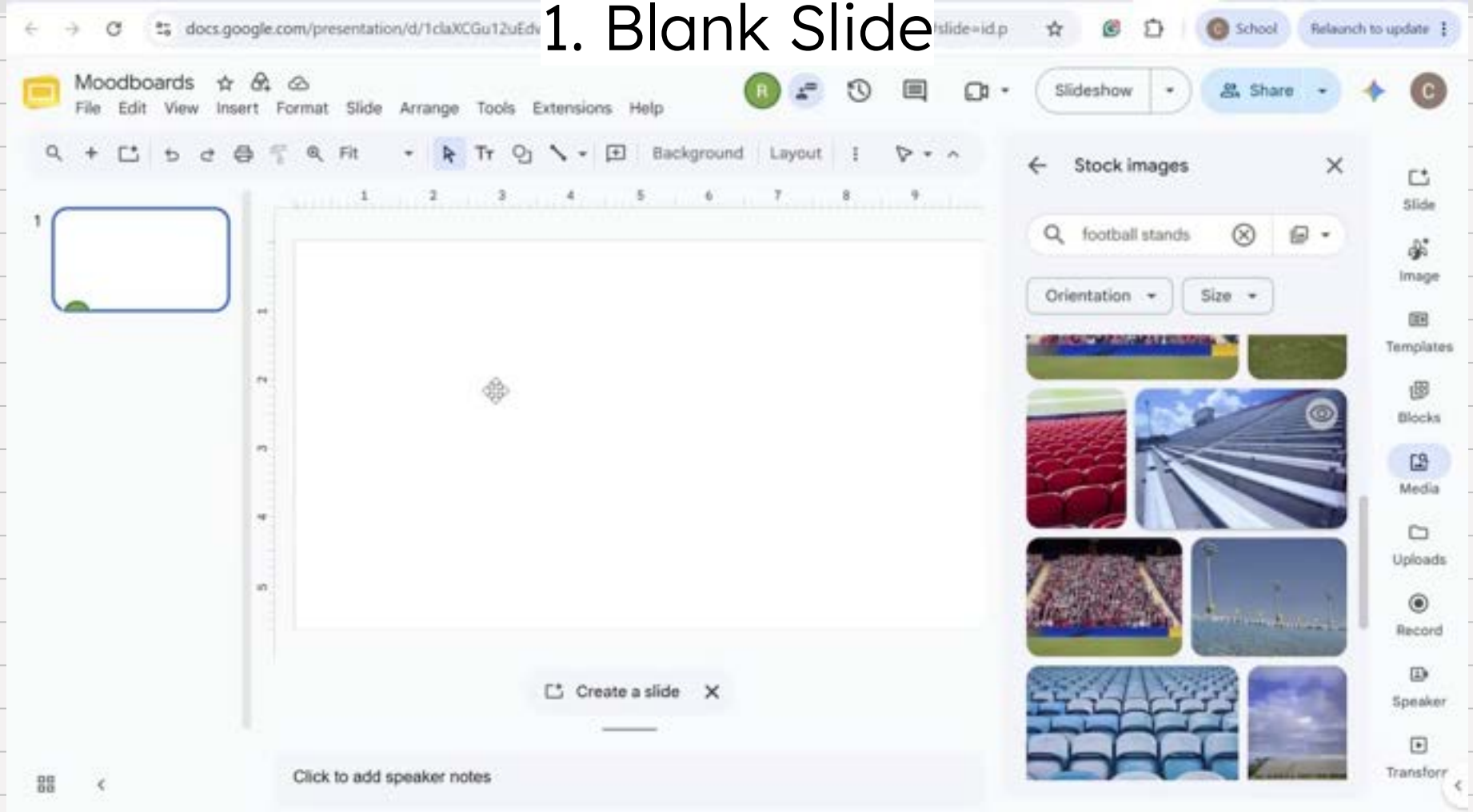
We can actively compare the user with solely visual versus multisensory stimulation and visually see how their inspiration changes within the moodboards

Participant: Claire

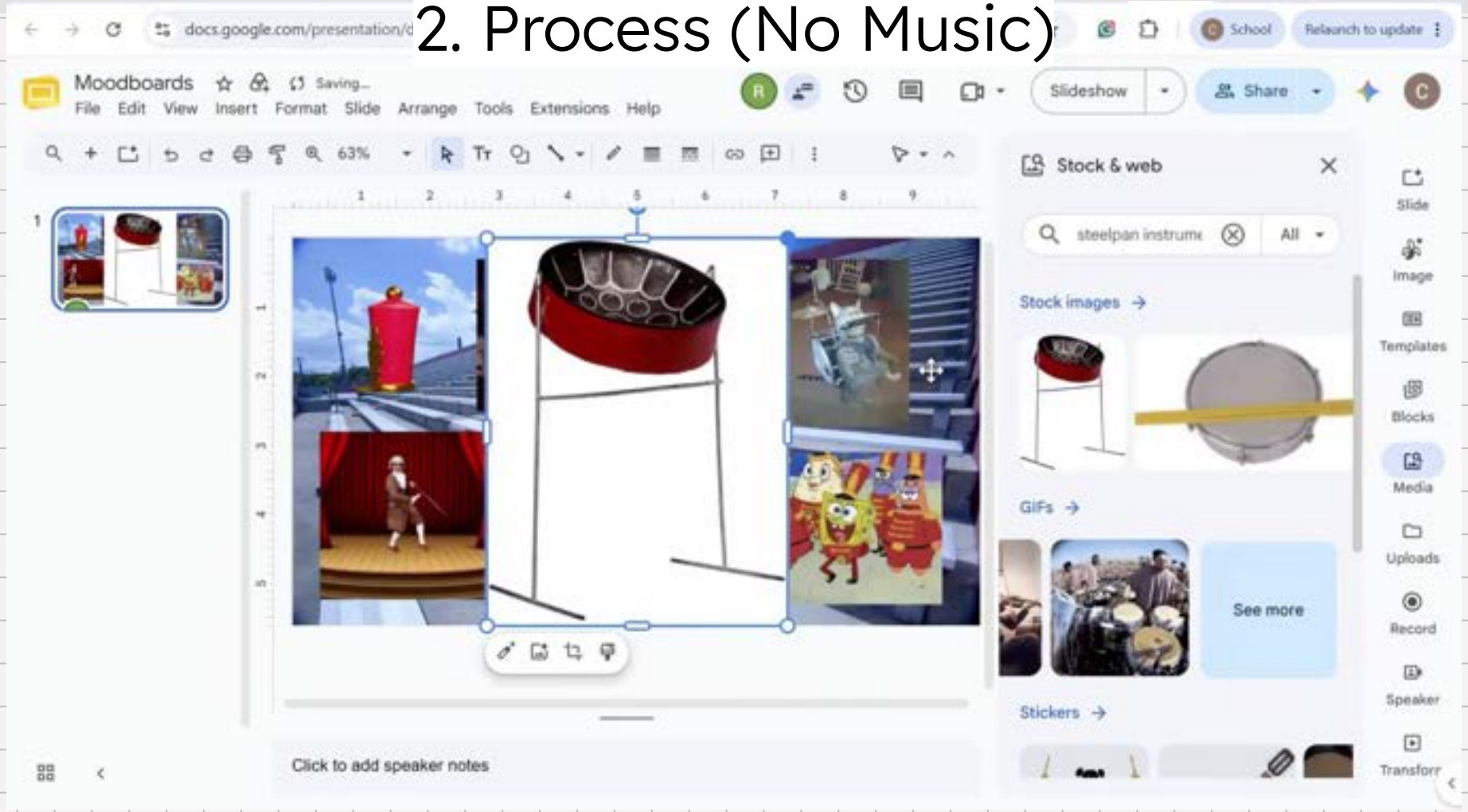


- **Stanford PubPol student with creative writing side hobby**
- **Interest in art (digital drawings)**
- **Past experience moodboarding for personal creative projects**

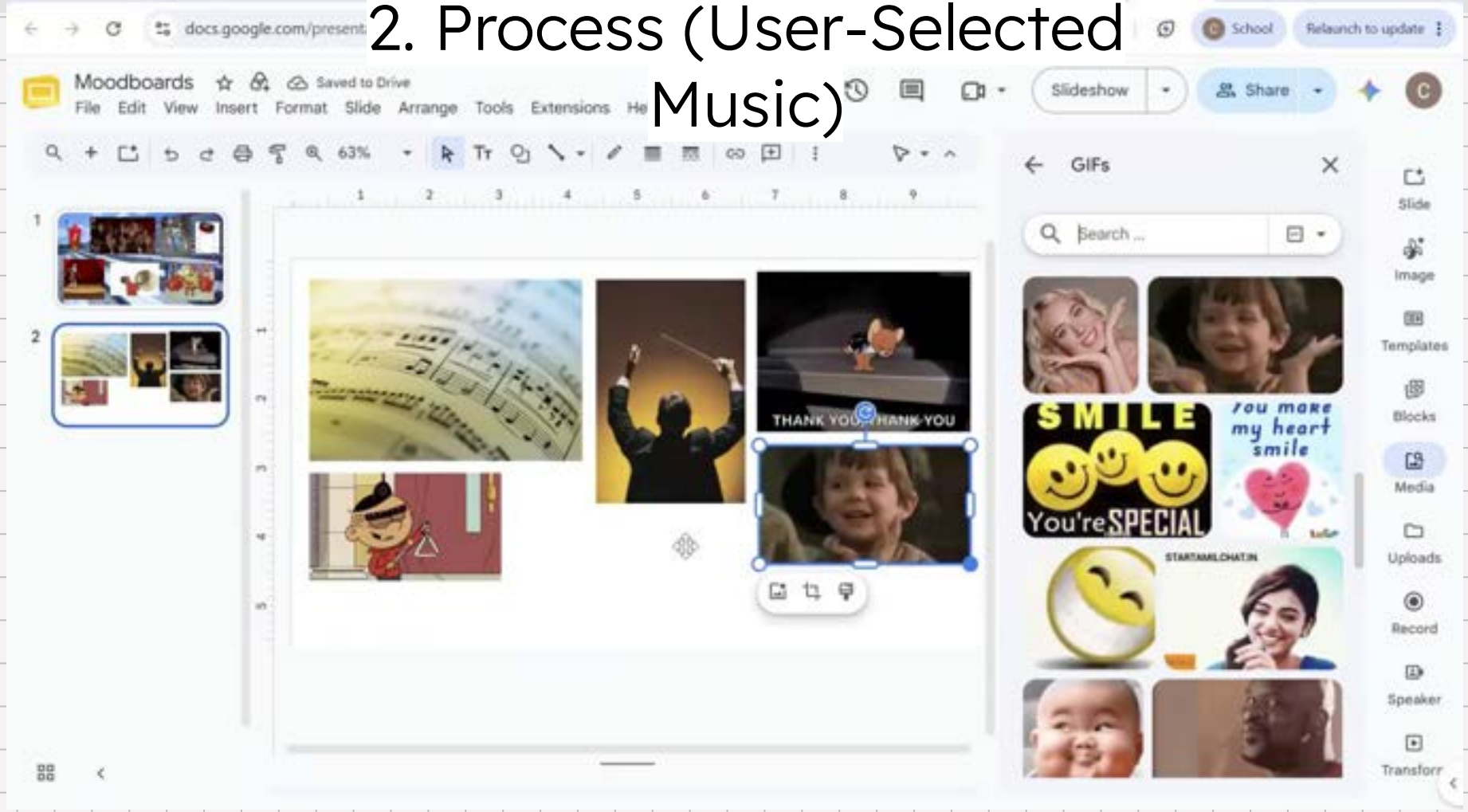
1. Blank Slide



2. Process (No Music)



2. Process (User-Selected Music)



3. 1st Moodboard, No Music









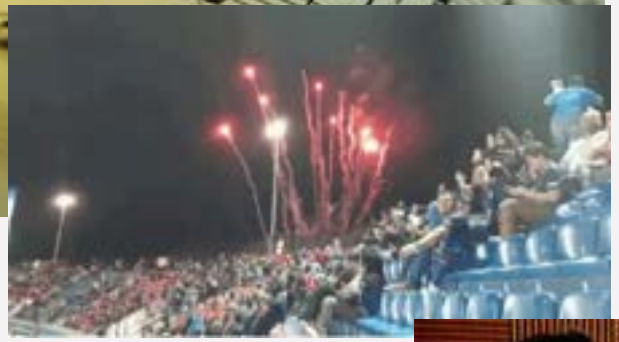






3. 2nd Moodboard, User-selected Music (Celtic Air and Dance Arr. Michael Sweeney)

















Assumption Reflections

Worked

- User felt more focused with music and noted feeling more immersed because the music brought her into her memories
- Moodboard had an overall clearer theme (classical music)

Didn't Work

- Immersion ≠ inspiration; immersion == focus
- Overall, both moodboards were very similar and drew from a specific memory in the participant's head

Insights

Did assumption hold?

Adding senses to a moodboard will make it more immersive

= Partially held

Insights:

1. **Held: sound deepened inspiration.** Literal → emotional, and she preferred it.
2. **Didn't hold: not a new experience.** It replayed her own band memories.
3. **Sound also steers:** the song chose the content (sheet music, a conductor).

Going forward: retest with an experience they've never had; choose the sound for them.

Name Department Course Number Section Time Location DATE

Creativity Piggybank



Prototype #3

Critical Assumption:
beginners are intimidated because they feel their work needs to be good enough to keep or show others, so hiding each piece until a quota is met lowers that intimidation

It's the premise for hiding work at all. If fear of showing isn't the barrier, hiding has no purpose.

Also tracked:

the eventual reveal pulls beginners back more than instant sharing

Experience Prototype #3

Setup:

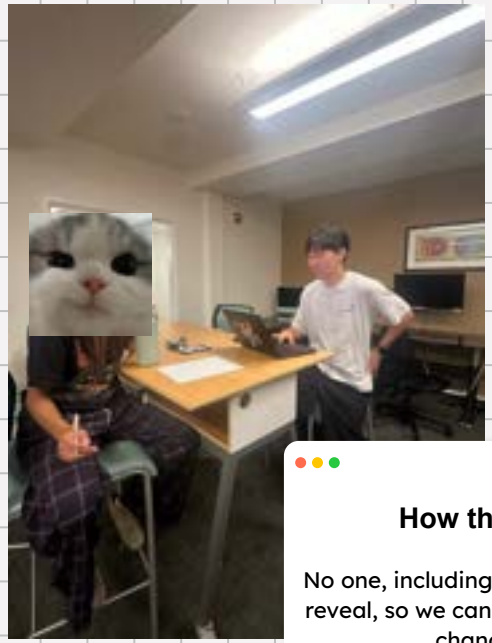
Draw the same prompt (your childhood room) three times, 2 minutes each.

Baseline: "how ready do you feel?" (1–5).

- 1: not confident in the drawing
- 5: confident in the drawing

After each drawing, the facilitator flips it face down into an envelope unseen; rating slip.

After the third, the participant opens the envelope, lays out all three, and decides keep, toss, or show.



How this tests the assumption:

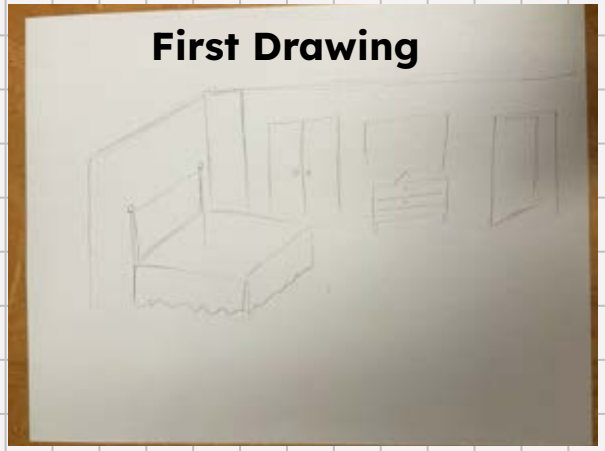
No one, including the participant, sees the work until the reveal, so we can watch whether removing the audience changes how they start and feel

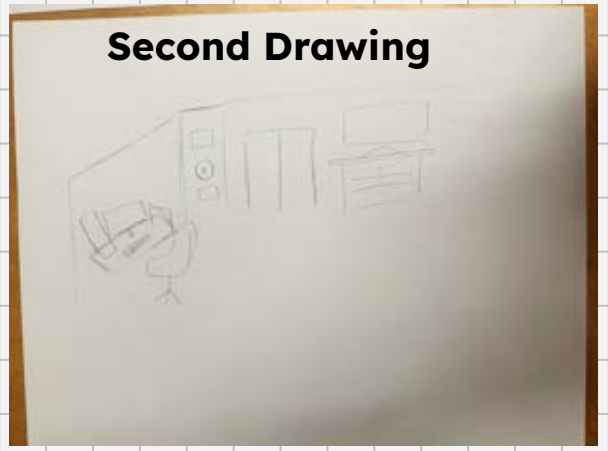
Participant: Evelynn

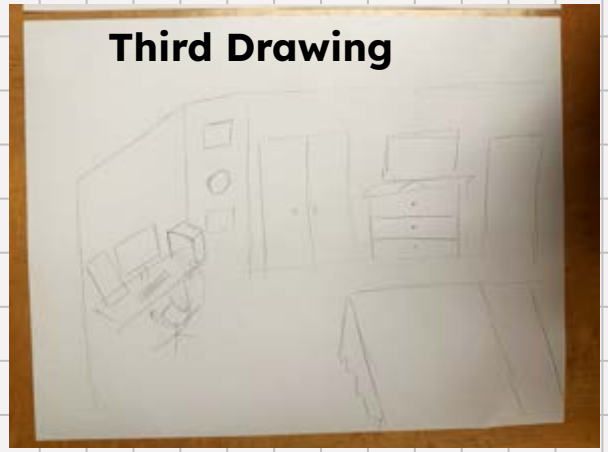


Evelynn

- Mechanical Engineering undergrad @ Stanford
- A beginner at drawing who wants to improve for her mechanical engineering classes.







	Baseline readiness	After Drawing 1	After Drawing 2	After Drawing 3
How did you feel drawing? Rating Confidence (1-5)	1	2	2	4

Assumption Reflections

What Worked

- **Confidence rose with repetition:** 1 → 4 by drawing 3, before the reveal
- **Improved without feedback:** bed "against the wrong wall" → "had everything in my room"
- **Reveal = pride:** "my last one I cooked"
- **Wanted more:** "would be nice if I got more than 2 minutes"

What Didn’t Work

- **Hiding didn't matter:** "didn't really care"
- **No audience to fear:** "I don't think I'd hang it up anyways"
- **Critic stayed:** "still don't like how it looks"
- **Prompt tested memory:** drew their room "from a while ago"

Insights

Did assumption hold?

Beginners are intimidated because their work must be good enough to keep or show, so hiding it lowers that.

= Did not hold

Insights:

1. **The critic is internal:** no audience to hide from. "I don't think I'd hang it up anyways"; "if not look good, crash out."
2. **Repetition builds confidence:** 1 → 4 by drawing 3, before the reveal.
3. **The reveal turns attempts into progress:** "my last one I cooked."

Going forward: keep repetition + reveal, drop hiding as the premise, test who sets the quota.

Moving Forward



We are moving forward with UGH-ly Hour

- Preserves what people value
- Addresses the time barrier
- Avoids making art an obligation
- Low effort to start
- Critical assumptions held

Ethical Implications

- “Silly prompts” can punch down rather than punch up
- Depending on what medium of creativity we pursue, certain visual or auditory impairments may hinder users from being able to participate

Target Demographic

Who it Serves

Busy workers and students whose creative practice and hobbies were lost under demanding workload and stress

Who it Leaves Out

- **People with motor or visual impairments for drawing prompts**
- **People where memes and drawing prompts don’t translate**
- **People who can’t afford to be distracted at work**

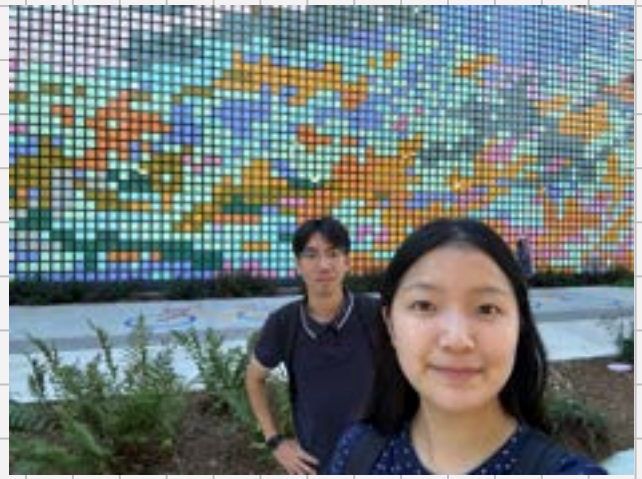
Appendix



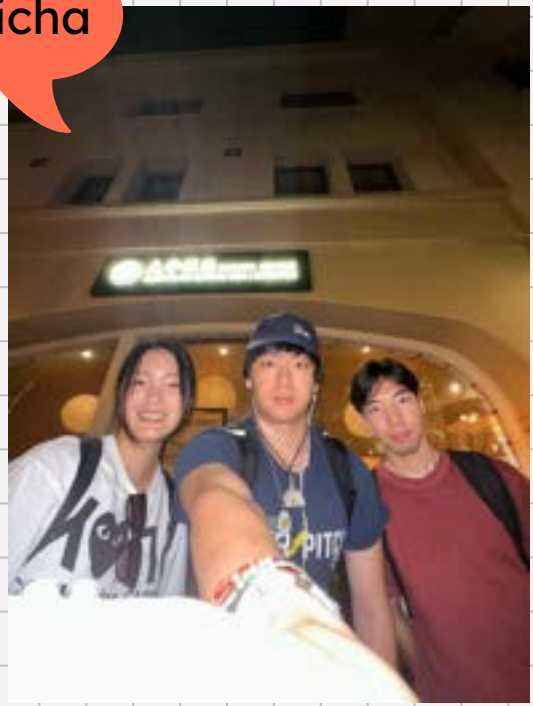
Interview Locations



Google,
Mountain View



Maruwu Seicha



Pov: Gary

We met Gary, a music theory and composition professor who has been involved in music for over 30 years, in his office

We were surprised that despite stressing the importance of routine and discipline in learning not only the reading of music but also the writing of music, he seemed rather neutral about generative AI in the music space.

We wonder that rather than being worried about AI "ruining" the art space, Gary is actually more confident in the work and performances humans produce, made by hours of discipline and kept routine, than in the lack of experience in listening to AI-generated music.

It would be game-changing if Gary could schedule and track set times per day or per week to work on his composition.

We met Joseph a third-year integrative biology student at UC Berkeley who previously attended an art charter school

We were surprised that Joseph did not want to study art because they believed that pursuing it full time would make it less appealing, and therefore they preferred balancing art with school

We wonder that Joseph, despite not having a lot of time for art, appreciates it more because it is an escape rather than an obligation, and they can be intentional about how they interact with it as a hobby

It would be game-changing if Joseph could plan out and manage time to maintain their relationship with art while engaging in harder courses and training to be a nurse

Pov: Joseph

We met Kevin, a creative writing professor who has a corporate background as a technical writer.

We were surprised that rather than short bouts of inspiration, Kevin values repetition and discipline and cites them for his creativity, where Kevin previously switched his role as a technical writer to part-time so he could spend more time writing

We wonder that for Kevin, the routine of engaging in creative writing replaces the discipline he sacrificed by choosing to pursue creative writing over technical writing and the idea of writing for production challenges the capacity of his personal creativity

It would be game-changing if there was a way to dedicate time to write non-work related poetry to maintain his creativity outside of his published works



We met with Alex, a Stanford alumnus who graduated with an MS degree in Computer Science & Graphics, but has a passion for painting and drawing.

We were surprised that despite finding value in the output of his artwork and being able to share his work with people he cares about, he finds the most meaning in the process and inspiration behind creation.

We wonder if Alex finds fulfillment in his work because he can say with confidence that it is something he made with his hands from a process that he himself has completed

It would be game-changing...

Pov: Alex

Pov: Edward

We met Edward, a fifth-year UBC computer science student and embedded firmware intern who taught himself digital art in high school by posting a drawing every day

We were surprised that his 100-day drawing challenge died at 40 once it became about engagement, and the drawing that survived is goalless notebook scribbling when “his mind’s melting,” which he says still isn’t the “way that he wanted”

We wonder if people want to watch themselves improve, but once that progress is measured by a streak or an audience, the practice breaks at the first busy season

It would be game-changing to let people watch their creative progress in a way that survives busy seasons, so a missed week never erases what they’ve built

HMW Methodology:

Team vote before the brainstorm.

Criteria: generates many kinds of solutions, what had ties to a surprise, must be distinct from the other two chosen HMWs.

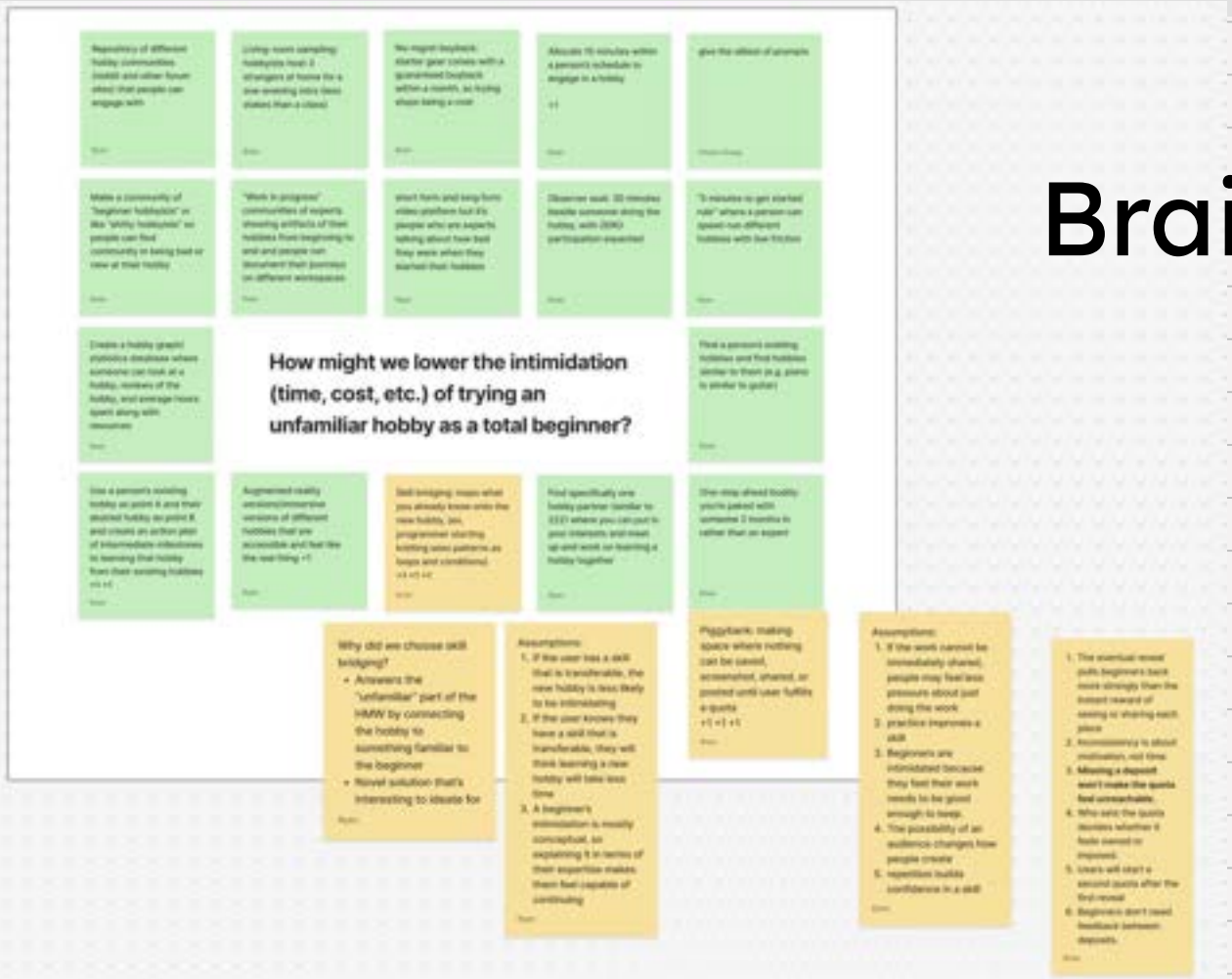
Solution Brainstorming



Solution Brainstorming



Solution Brainstorming



Assumption Reflections 1

What Worked

- **The silly prompt lowered self-judgment:** rated judging themselves 2/5 on the meme vs. 4/5 on the self-portrait
- **Low stakes didn't make it throwaway:** meme drawing"mattered" 4 vs. 2 for the portrait. The dog went in KEEP; the portrait went in TOSS.
- **Overall more fun:** "I had more fun doing the dog"

- **A silly prompt gives more ways in:** pointed to the "variety in memes": almost any meme was a valid answer, so starting was easy.
- **The time cap gave permission to stop:** giving time limit made it “more approachable." Without a limit, "felt more pressure to make it look better," because "I like to see something finished."
- **The break itself landed mid-work.** They "actually had fun doing both."

Assumption Reflections 1

What Didn't Work

- **The serious prompt → blank-page freeze.** "What would a creative self portrait be?" They didn't plan where to start, and the portrait came out "less human-like." An open, serious prompt read as a test.
- **3 minutes felt long** for a break in the middle of work.
- **The comparison mixed two variables.** Self-portrait vs. meme changed the subject and how much choice they had, not only the silliness, so "variety in memes" may explain part of the gap. Next run: keep the subject fixed and change only the framing.

Assumption was **supported** by our observations

- **Surprise:** the silly drawing was judged less *and* mattered more. Low stakes didn't make it throwaway.
- **Going forward:** concrete, silly prompts with many valid answers; a time box under 3 minutes; keep/toss with no audience.

Name Department Course Number Section Time Location DATE

Assumption Reflections 3

Worked

- **Confidence climbed with repetition before reveals.** Readiness started at 1/5. Ratings went 2, 2, then 4 before they had seen any drawing again.

- **Each round got better without feedback.**
  - Drawing 1: "I don't like the perspective I drew it from… bed against the wrong wall."
  - Drawing 2: switched focus to the desk and liked it more.
  - Drawing 3: "had everything in my room."

- **The reveal made progress visible.** Laying out all three, they laughed, winced, and compared, then said "my last one I cooked."
- **Repetition built a mental map.** "I started remembering where everything was. You get a good gauge of laying it out."
- **They wanted more.** "If I had more time, [I'd have] drawn the squishmallow in the corner"; "would be nice if I got more than 2 minutes."

Assumption Reflections 3

Didn't Work

- **Hiding the drawings had no felt effect.** They "didn't really care" about not seeing the earlier ones.
- **There was no audience to fear.** "I don't think I'd hang it up anyways." The barrier is their own judgment. Asked what stops them practicing: "If [it does]n't look good, crash out."
- **The prompt tested memory as much as drawing.** For drawing 1 they "went too far back" and drew an old version of their room. They also stuck to objects: "I don't like drawing living things."

Assumption partially invalid

Invalid: hiding work from others changed nothing; there was no audience to fear

Valid, unexpected: the reveal added pride ("my last one I cooked"). Confidence came from seeing improvement, not from hiding.

Going forward: keep the reveal as a progress moment and drop "no audience" as the premise.